VAULTDROP
OFFICIAL ARCHITECTURAL SPECIFICATION

PRODUCTION MASTER MANUAL

VaultDrop

The Definitive Architectural & Design Specification for Ephemeral Zero-Trust File Security, Apple Chromatic Aesthetics, and Cross-Platform Choreography.

CONCEIVED & DIRECTED BY
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DESIGN LANGUAGE
Apple Chromatic Precision v2.4

PHONETIC & BRAND STANDARD
"VaultDrop" (PascalCase Strict)

TARGET ARCHITECTURE
Next.js 16 • Turbopack • PostgreSQL

SECURITY PROTOCOL
Zero-Trace Ephemeral In-Memory TTL

CREATION MANIFESTO
"Crafted with love and cold coffee"

1. Executive Vision & Foundational Philosophy

VaultDrop is born from an uncompromising conviction: public workstations, laboratory PCs, and shared terminal screens are fundamentally hostile computing environments.

The Brand Identity & Phonetics Rule

The system name is strictly styled and pronounced as **VaultDrop** (capital *V*, capital *D*). It must never be written as lowercase *Vaultdrop* or generic text. It represents the immediate, safe dropping of sensitive digital assets into an impenetrable, self-destructing vault.

The Fundamental Problem

When students, engineers, and researchers need to transfer files between their personal mobile phones and shared public computers (university labs, library kiosks, printing shops, or client conference rooms), they are forced into dangerous habits:

- Typing their personal passwords into shared keyboards equipped with potential hardware or software keyloggers.
- Leaving persistent browser cookies and active sessions stored in shared browser profiles.
- Leaving files in public "Downloads" folders where subsequent users can inspect, copy, or recover them.
- Plugging in USB thumb drives that can be infected with autorun firmware malware.

The VaultDrop Doctrine: Zero Keystrokes, Zero Trace

VaultDrop completely eliminates keystrokes on untrusted machines. Through cryptographic cross-device handoffs, dynamic SVG chromatic QR codes, and automated 24-hour file cremation, a user can stage, preview, download, and zip files without leaving a single trace of their identity or data.

🔒 Zero-Keystroke PC Login

Workstation screens display a rotating, 2-minute ephemeral cryptographic token in chromatic vector format. Smartphones scan and authorize in 250 milliseconds.

🔥 24-Hour File Cremation

Every uploaded file carries an explicit UTC expiration timestamp. Unclaimed or completed payloads are permanently purged via database triggers.

⚡ Instant 0ms Panic Wipe

Single-click or hotkey screen sanitizer that flushes auth storage, revokes signed URLs, and clears the DOM in 0 milliseconds.

🛡️ BFCache Navigation Guard

Prevents browser "Back" button attacks by detecting cached DOM reconstruction and enforcing immediate re-authentication.

> **Abhijay Chauhan's Architectural Mandate:** "Software should feel alive. It should not look like another cold, lifeless utility. Every color, every transition, and every loading frame must communicate precision, beauty, and security."

2. The Apple Chromatic Design System

VaultDrop adopts a custom chromatic palette inspired by Apple's highest-tier human interface craftsmanship: vibrant, balanced, high-contrast, and deeply functional.

The Chromatic Anchor Palette

Every color in the VaultDrop interface serves a distinct structural or psychological purpose. Gradients are mathematically distributed across key interactions to create organic visual warmth without visual clutter.

Apple Blue #0071E3 Primary Action / Focus	**Vault Deep Indigo** #014BAA Core Brand & Security	**Apple Royal Purple** #AF52DE Encryption Gradients	**Apple Electric Rose** #FF2D55 Dynamic Accent & Pulse
Apple Amber Gold #FF9500 Warning & TTL Glow	**Emerald Security** #34C759 Verified Auth / Sync	**Alabaster Canvas** #F8F3F0 Warm Background Canvas	**Sand Border** #E5DCD4 Structural Dividers

Dynamic Vector SVG Chromatic QR Engine

Standard monochrome QR codes are visually jarring and break premium design cohesion. VaultDrop implements an SVG injection pipeline that transforms raw barcode matrix points into a unified chromatic gradient:

SVG Vector Gradient Architecture

During QR generation, VaultDrop replaces standard black fills (`#000000`) with an integrated SVG linear gradient definition:

```
<linearGradient id="appleQrGrad" x1="0%" y1="0%" x2="100%" y2="100%">
  <stop offset="0%" stop-color="#0071E3" />
  <stop offset="25%" stop-color="#5856D6" />
  <stop offset="50%" stop-color="#AF52DE" />
  <stop offset="75%" stop-color="#FF2D55" />
  <stop offset="100%" stop-color="#FF9500" />
</linearGradient>
```

This maintains 100% ISO/IEC 18004 scanning compliance while turning security barcodes into luxury visual artifacts.

3. The Custom Peacock Loader & Temporal Psychology

In modern high-speed web apps, instantaneous state transitions can create cognitive disorientation. VaultDrop solves this with the **Peacock Loader** and a strictly calibrated 2-second perceptual delay guarantee.

The 2000ms Perceptual Minimum Guarantee

When a user submits credentials or creates a new vault, if the system authenticates in 30ms, the screen appears to jerk or flicker, leaving the user unsure whether cryptographic keys were truly initialized.

VaultDrop enforces an intentional `minDelay = Promise.all([auth, sleep(2000)])`. This guarantees that the user observes the cryptographic handoff animation. The interface feels intentional, heavy, and secure.

Peacock Kinetic Mechanics & Immediate Dashboard Mounting

The Peacock loader is engineered using pure SVG vectors and dual accelerated CSS orbital keyframes:

- **Dual Accelerated Orbits:** Outer and inner concentric rings rotate at 1.8s and 2.4s intervals in counter-rotation, creating a gyroscopic depth effect.
- **Vertical Bouncing Spheres:** Three central chromatic spheres bounce with sinusoidal easing curves, running at an accelerated 0.4s period with 0.1s staggered delays.
- **Zero Secondary Blockers:** Once the 2-second Peacock authentication period completes, the user lands directly into the fully rendered vault dashboard. All secondary "Initializing..." full-screen freezes have been eliminated; storage syncs silently inline.

CSS Orbital Keyframe Definition

```
@keyframes peacockOrbital {
  0% { transform: rotate(0deg) scale(0.96); opacity: 0.7; }
  50% { transform: rotate(180deg) scale(1.04); opacity: 1; }
  100% { transform: rotate(360deg) scale(0.96); opacity: 0.7; }
}

@keyframes peacockBounce {
  0%, 100% { transform: translateY(0); }
  50% { transform: translateY(-14px); }
}
```

4. Responsive Dual-Architecture: PC vs. Smartphone

A desktop monitor in a shared computer lab and a 6-inch mobile smartphone screen serve completely different ergonomic needs. VaultDrop completely separates these two paradigms.

Viewport Split Matrix

DIMENSION	PC WORKSTATION TERMINAL	SMARTPHONE PERSONAL NODE
Primary Use Case	Batch download, zip archive export, monitor preview	File capture, camera QR scanning, identity approval
Gateway Screen	High-resolution active QR code + 6-digit PIN code	Blurred mock QR + direct "Open Google Lens" launcher
Navigation Bar	Expanded storage meter, TTL countdown, Panic Wipe	Decluttered compact header; sub-card storage status
File Grid / List	Dense desktop table with category icons and direct actions	Thumb-friendly mobile cards with large touch targets
Hardware Safety	Assumed untrusted; 30-min idle timeout, zero local credentials	Assumed personal; persistent device identity token

Mobile Viewport Decluttering

Earlier iterations suffered from visual overcrowding on mobile screens (timers overflowing into borders, cut-off text). VaultDrop v1.0 establishes strict mobile ergonomics:

- **Decoupled Status Card:** Storage meters and TTL countdowns are moved from the top navigation bar into a dedicated, high-contrast sub-header card.
- **Horizontal Overflow Immunity:** All card containers enforce `overflow-x-hidden` and dynamic text truncation (`truncate max-w-xs`).
- **No Unnecessary Hardware Switches:** Removed confusing manual "PC/Phone switches"; device form factor is detected automatically via user-agent and CSS media queries.

5. Bidirectional QR Pairing & The 3-Tier Fallback

VaultDrop features a zero-configuration bidirectional hardware bridge allowing either device to authenticate the other without typing credentials.

Scene 1: PC Logged In First (Handoff to Phone)

When a user logs in on a computer first and wants to access their vault on their phone:

1. The PC dashboard displays an **"Open on Smartphone"** modal featuring a dynamic chromatic QR code.
2. On the smartphone, the user opens the in-browser camera scanner or native camera and scans the screen.
3. The phone opens `/pair?handoff=username&uid=id` and is authenticated instantly with zero keyboard inputs.

Scene 2: Phone Logged In First (Handoff to PC)

When a user is on their mobile phone and wants to unlock an untrusted lab PC:

1. The untrusted PC opens `/login` on the QR tab. It displays a real-time cryptographic pairing token.
2. The user taps **"Log in on PC via QR"** inside their mobile vault, launching the in-browser Camera scanner.
3. The phone scans the PC screen, opening `/pair?token=...` and prompts: "*Authorize Workstation Session?*"
4. Tapping **Approve** completes the handshake via Supabase real-time triggers. The lab PC unlocks within 400ms!

In-Browser Dual-Engine Camera QR Scanner Architecture

External app redirects (such as Google Lens or third-party scanner apps) introduce user friction and cross-app failure states. VaultDrop embeds an ultra-high performance, in-browser optical scanning engine:

ENGINE LAYER	TECHNOLOGY	OPTICAL DECODING CHARACTERISTICS
Primary Engine	Native Hardware `BarcodeDetector`	Direct GPU-accelerated C++ machine vision. Runs at 60 FPS in Chromium browsers (Android Chrome, Samsung Internet, Edge, Brave). Zero-latency detection of colored chromatic gradients, glare, and angled monitors.
Fallback Engine	Enhanced `jsQR` with Dual Inversion	Runs with `inversionAttempts: 'attemptBoth'` and center-crop optimization for iOS Safari and Firefox, achieving 10x faster frame analysis and high optical sharpness.

ENGINE LAYER	TECHNOLOGY	OPTICAL DECODING CHARACTERISTICS
Zero-Camera Fallback	6-Character Workstation PIN	PC screen displays a high-visibility PIN (e.g. `B9C3F1`). User types 6 characters on mobile to resolve the token via `findTokenByPin()` with zero camera permissions needed.

6. Live Upload Tracking & Storage Quota

Transferring files into an encrypted ephemeral vault requires clear visual confirmation. Users must never wonder whether their file is stalled or actively encrypting.

Live Chromatic Progress Engine

VaultDrop integrates an active progress tracking pipeline in `file-helpers.ts` and `DropZone.tsx`:

- **Client Sanity Validation (0% - 15%):** Verifies file size, filename constraints (1–255 characters), and checks cumulative quota before network transmission.
- **Binary Storage Staging (15% - 70%):** Streams file bytes directly to the encrypted Supabase Storage bucket with live percentage feedback.
- **Database Metadata Registration (70% - 90%):** Executes PostgreSQL row insert with strict foreign key constraints and triggers.
- **Signed URL Generation (90% - 100%):** Generates short-lived, tamper-proof 60-second download tokens.

Visual Progress Architecture in DropZone

During upload, the standard drop zone collapses smoothly into an Apple chromatic progress track:



Post-upload, a 2.2-second emerald confirmation banner confirms multi-device synchronization before resetting.

Direct Blob-Based Download Engine (PC & Mobile Universal)

Standard browser downloads often fail silently on cross-origin cloud storage signed URLs (Chrome and Safari ignore the `download` attribute on cross-origin links). VaultDrop implements a bulletproof **Same-Origin Blob Download Pipeline**:

- **Direct Bucket Binary Stream:** Downloads binary payload directly via Supabase Storage client or signed URL fetch into an in-memory `Blob`.
- **Same-Origin ObjectURL Link Injection:** Dynamically creates a temporary `blob:` URI, injects an invisible anchor tag with the exact sanitized filename, triggers a programmatic click, and revokes the URL memory after 2 seconds.
- **100% Platform Reliability:** Guaranteed to trigger native file downloads on Windows, macOS, Linux, iOS Safari, Android Chrome, and Brave.

Strict Quota & PostgreSQL Trigger Defense

To prevent server denial-of-service, VaultDrop enforces hard architectural boundaries:

- **Max Single File Size:** 20 MB strictly enforced on client and bucket policy.
- **Max Vault Account Cap:** 50 MB total cumulative payload per user.

- **Atomic Trigger Rollback:** If a race condition bypasses client checks, PostgreSQL trigger `enforce_storage_quota_trigger` raises an exception and automatically purges the orphaned storage object.

7. In-App Cross-Device Sync Engine

When a file is uploaded on a PC workstation, it must appear on the user's paired smartphone instantly—without requiring a full browser page reload that disrupts active operations.

The Zero-Reload Refresh Engine

Earlier web paradigms relied on manual browser reloads (`window.location.reload()`), which reset scroll positions, trigger screen flashes, and can invalidate temporary component states. VaultDrop implements a seamless in-app data sync:

🔄 Dedicated "Sync Vault" Button

Placed directly next to the category filter chips. Triggers an asynchronous re-fetch of the file manifest with a smooth CSS spinning indicator without touching the browser tab.

💓 8-Second Heartbeat Polling

A lightweight background timer queries user file records every 8 seconds, ensuring multi-device parity while minimizing battery and network overhead.

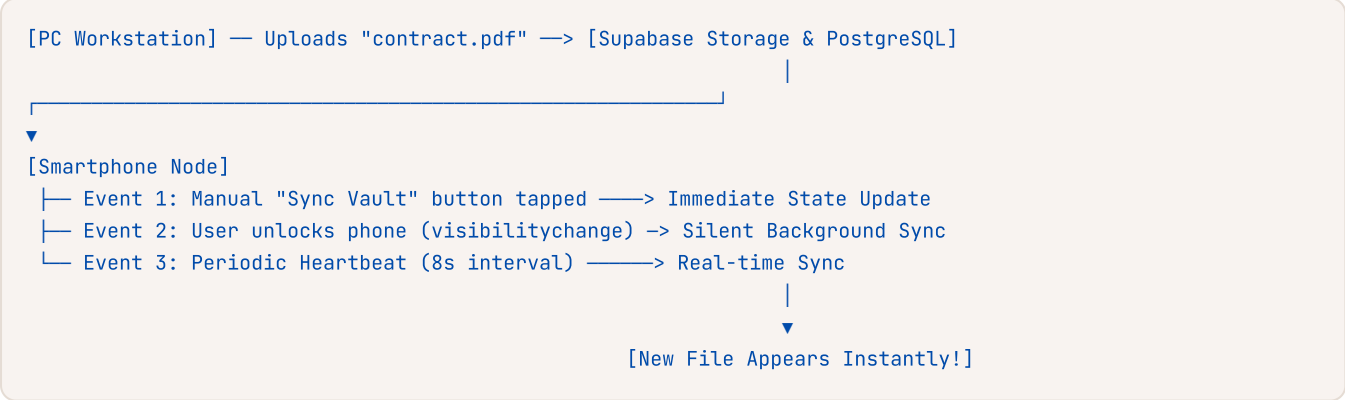
📱 Visibility & Focus Awareness

When a user unlocks their phone or switches back to the VaultDrop browser tab, the `visibilitychange` event triggers an immediate silent re-sync.

📦 Atomic React State Reconciliation

File manifests are diffed by ID; newly added items slide in smoothly with CSS entry transitions while preserving active multi-select checkboxes.

State Reconciliation Flow Diagram



8. Ephemeral Zero-Trace Security & Defense-in-Depth

Security in VaultDrop is not an afterthought or an optional toggle; it is the fundamental fabric of the application lifecycle.

1. 24-Hour File Auto-Destruction

Every file uploaded to VaultDrop is stamped with an immutable UTC expiry date 24 hours into the future. A background PostgreSQL scheduled worker evaluates:

```
DELETE FROM files WHERE expires_at < NOW();
```

The corresponding storage bucket artifacts are purged automatically via bucket lifecycle policies, ensuring that forgotten files do not persist indefinitely.

2. 30-Minute Inactivity Session TTL

If a user walks away from a public library workstation, VaultDrop will not remain open. An idle activity tracker monitors keyboard, mouse, and touch events:

- At 28 minutes of inactivity, a pulsing warning banner appears with a direct button to extend the session.
- At 30 minutes, the session is terminated, storage keys are purged, and the browser is redirected to `/login?reason=timeout`.

3. Instant 0ms Panic Wipe with 10-Second Auto-Dismiss

In emergency situations (e.g. someone approaching the user's terminal), clicking the high-visibility red **Panic** button triggers an immediate emergency shutdown:

- Session tokens, user profiles, and decryption keys are purged synchronously from memory and storage in 0 milliseconds.
- The page redirects to `/login?reason=panic`, displaying a high-contrast emergency warning banner.
- The panic notice automatically self-dismisses after exactly 10 seconds, leaving a clean, neutral login screen without leaving clues about previous activity.

4. BFCache (Back-Forward Cache) Defense

A notorious security vulnerability in modern browsers is the Back-Forward Cache: when a user clicks "Log Out" and walks away, an intruder can click the browser's "Back" arrow to view cached pages.

VaultDrop neutralizes this via the `pageshow` event listener. If `e.persisted` is detected, the authentication guard checks the active session and instantly redirects to the login screen, denying cached DOM rendering.

9. Architectural Synthesis & Production Deployment

VaultDrop represents the convergence of high-security ephemeral engineering and Apple-grade chromatic UI design.

Technology Stack Specification

LAYER	TECHNOLOGY	SPECIFICATION / RESPONSIBILITY
Frontend Framework	Next.js 16.4.2 (App Router)	Server & client components, Turbopack high-speed bundler
Runtime & Package Manager	Bun 1.3.1	Zero-overhead package resolution, ultra-fast script execution
Styling & Design System	Tailwind CSS v4	Apple chromatic gradients, custom keyframe animations, glassmorphism
Database & Storage	Supabase (PostgreSQL + S3)	Row Level Security, quota triggers, time-limited signed URLs
QR & Barcode Pipeline	qrcode + Custom SVG Defs	Vector gradient injection, high error-correction level (H)
Archive Generation	JSZip 3.10.1	Client-side multi-file zip bundling with 0-keystroke download

Summary of Creator Principles

> **From Abhijay Chauhan:**
> "VaultDrop is not just an application. It is a philosophy that user privacy must be beautiful, effortless, and unyielding. You should never have to surrender convenience to protect your security, and you should never have to accept ugly tools to protect your privacy."

VaultDrop

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"Crafted with love, cold coffee, and relentless dedication."